



PCB Design Review Checklist

 Mark Harris | Created: October 25, 2021 | Updated: August 10, 2026

At a Glance

It's essential to check a board for issues before production or prototyping. Here's an extensive checklist you can adapt or use directly for PCB design review.

[Read Full Summary](#) 

 Go Deeper with AI:



At the various companies and organizations I've worked for, there have been a wide variety of procedures for reviewing and approving a [PCB design](#) for production. It has been more of a quick look over the board at some of the smaller companies to see if there are any glaringly obvious mistakes, and maybe, if the reviewer remembers, running a design rule check report. Even at larger organizations, there has been a distinct lack of procedure and consistency involved, often because there was simply never enough time to create a procedure with the pace of work and the workload levels of all the relevant staff.

We all know how important it is to check a board for issues before proceeding to production or prototyping. No matter how much effort you put into the design, inevitably, something slips through the gaps. If you do not have a robust list of items to look at before sending off your design files, those issues might still slip through to the board fabricator or assembler.

As we're always operating under such high workloads and pressures to get things done, it can be hard to find the time to generate a procedure or process for reviewing your board's design, as I mentioned above. So, rather than talking in this article about the evils of not checking board designs over, instead, I'm going to make your life easier by providing an extensive PCB checklist you can either adapt or use directly for reviewing each board design. Depending on your board and the technologies used on it, some items on this list may not be relevant - or there may be items missing. While this list is relatively extensive, it's not exhaustive as I've intended it to be the starting point for your own process. Hopefully, saving you from taking hours to come up with your own PCB quality



Solutions for Every Role and Team

- Discover and evaluate design options earlier
- Develop in one shared workspace
- Scale with control and compliance
- Connect every stage of hardware development

[Learn More](#)



Watch on YouTube

Before Submitting Your Board for PCB Design Review

- Check for unrouted nets
- Repour all polygons
- Ensure your board has a complete silkscreen, including:
 - Company logo(s)
 - Product logo(s)
 - Copyright notice
 - Warning/Hazard label(s) and Icons
 - Connectors are labeled, and pinouts are called out where relevant
 - QA/Test block with barcode, blank areas for a serial number, dates, and QA/Test checkmarks
 - Board name, print date, and revision number
 - If there are designators on the silkscreen, then check:
 - Each designator is close to and clearly identifies the associated component
 - All designators are only in one or two orientations
 - Text size and font will remain legible after the fabrications process
 - ICs have pin one clearly marked, i.e., the pin one marker is not located under another component
- Update the [PCB from its schematics](#) to ensure the schematic and board are synchronized
- Ensure the design rule report passes with no errors
 - Ensure there is a design rule to catch nets with only 1 pin
- Ensure you have the board outline on a mechanical layer that will go to the board fabricator
- [Fiducials](#) are present for the assembly and meet these criteria:
 - Three board-level fiducials are included as a minimum
 - Two fiducials are diagonally opposing each other across all extremely fine-pitch components
- Mounting points have sufficient clearance for the chosen washer and screw head
- If an enclosure model is available, it has been tested against the board to ensure there will be no interference between the enclosure and components
 - All components (including mechanical items) have accurate 3D models

Your board should be fully production-ready, as far as you are concerned, before sending it to colleagues for a design review checklist to avoid any nugatory effort on their part.



Solutions for Every Role and Team

- Discover and evaluate design options earlier
- Develop in one shared workspace
- Scale with control and compliance
- Connect every stage of hardware development

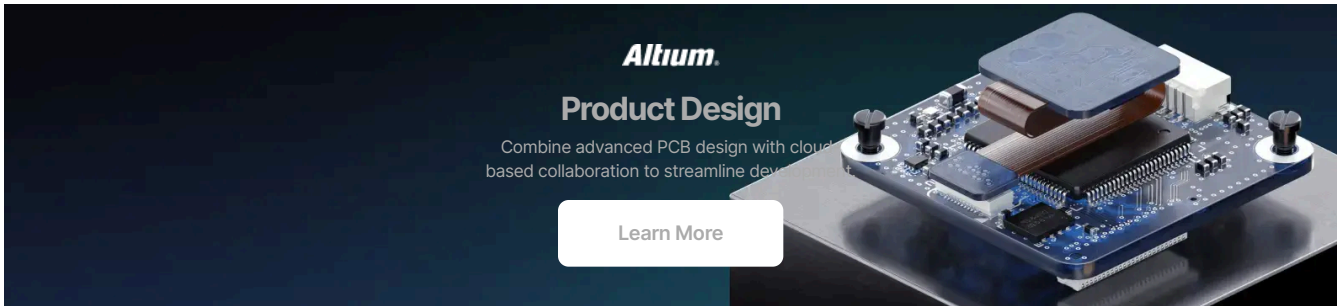
[Learn More](#)

Layer	Thickness	Material	Weight	Thickness
Layer - 5	0.0210	0.0210 (3-7628)	370H	4.34 / 0.0180
	0.0012	2113 - 59%	1oz P/G	
	0.0069	2113 - 59%	370H	3.99 / 0.0220
Layer - 6	0.0006	0.0045 (1-2116)	1/2oz Sig	
	0.0045	2113 - 59%	370H	4.14 / 0.0200
Layer - 7	0.0012	0.0036	1oz P/G	
	0.0036	0.0020	370H	3.99 / 0.0220
Layer - 8	0.0020	0.0010	1/2oz Sig (Std Plt)	
			Taiyo 4000-HFX DI	3.50 / 0.0190

|| Through

Materials: Isola 370H Isola370HR High-Tg FR4

Example stackup table for comparison in a PCB design review. Image provided by [Zach Peterson/TTM](#).



Altium
Product Design
Combine advanced PCB design with cloud-based collaboration to streamline development.
[Learn More](#)

Signal Path

- Ensure that the ground plane has sufficient current-carrying vias near connectors, and voltage/return sinks
- If required/relevant, ensure voltage planes and areas have sufficient connection vias for current requirements
- Ensure tracks to reference planes are sufficiently wide for current requirements
- Check there are sufficient quantities of vias for the current-carrying capacity of traces
- Check the minimum track widths for all current-carrying nets are sufficient. Consider if a design rule is required
- Check all ground pins have a via to the ground plane
- Check there is a continuous ground plane within one signal layer of any signal trace
- Check any [controlled impedance](#) traces have the correct net rules and impedance profile
- Check differential pair tracks are as close together as possible
- Check differential pair tracks lengths are matched
- Check all high-speed signals are length matched, specifically including:
 - DDR
 - PCIe
 - Ethernet
 - LVDS



Solutions for Every Role and Team

- Discover and evaluate design options earlier
- Develop in one shared workspace
- Scale with control and compliance
- Connect every stage of hardware development

[Learn More](#)

The xSignals package has features in it that can help automate many of these checks after the PCB layout is finished, but before a more rigorous manual review by the engineering team.



[Return path violation](#) highlighted in xSignals. You can choose to ignore this or re-do portions of the routing to eliminate it.

Components

- Check all through-hole pads are set to plated if they are going to be soldered
- Check there is sufficient clearance for the following:
 - For pick and place heads in production
 - For hand assembly in prototypes
 - For soldering pen tip access if any rework is required
- Check bypass capacitors are placed as close to IC power pins as possible (under 15mm)
- Check crystal/oscillator clock sources are as close to the IC clock pins as possible
- Check termination resistors are as close to the signal source as possible
- Check EMI/RFI filtering is as close to the exit point (board edge, connector, shield) as possible
- Check potentiometers increase signal/voltage when turned clockwise
- Check programmable devices have accessible programming header/pads
- Check no high thermal mass components (e.g., large transformers/inductors) are located next to very small components
- Check component placement prioritizes short track lengths for high-speed signals
- Check there are a sufficiently large copper area for heat-sinking high-dissipation devices, including:
 - Linear regulators
 - Switched-mode power supplies (including LED drivers)
 - High power LEDs
 - High-frequency gate drivers
 - MOSFETs
 - Motor drivers
 - Chargers
 - High-speed microprocessors
 - Power amplifiers

Testing

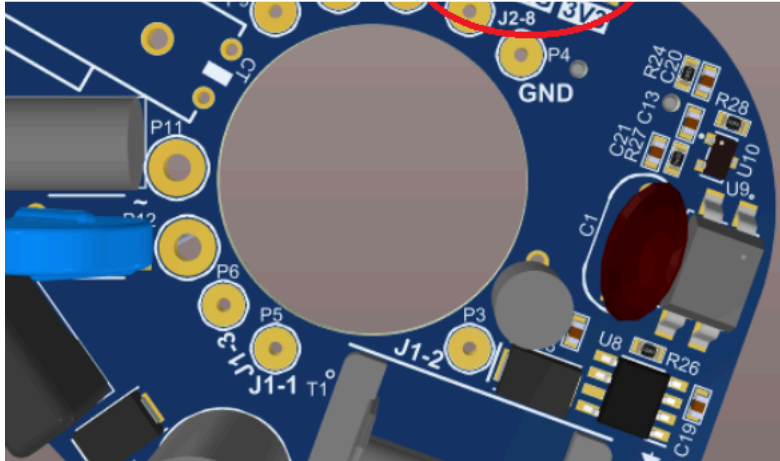
- Check that test pads are sufficiently far from the edge of the board to allow fixturing
- Check test pads do not create stubs/impedance mismatch on high-speed nets
- Check that component do not obstruct access to test pads for manual testing (e.g., using an oscilloscope probe) or automated testing (e.g., using a bed of nails)
- Check test pads are clearly labeled for prototypes
- Check any signal needed for testing or inspection has a test point



Solutions for Every Role and Team

- Discover and evaluate design options earlier
- Develop in one shared workspace
- Scale with control and compliance
- Connect every stage of hardware development

[Learn More](#)



These [Keystone testpoints](#) are placed very close to the board edge, and their silkscreen markings are placed very close to other through-hole pads for wire soldering. If there is no spec that requires these testpoints to be here, a new location could be identified in a PCB design review.

Protection/EMI/EMC

- Check appropriate creepage and clearance rules are set for all high-voltage nets
- Check if separate earth tracks/paths are necessary for ESD
- Check there are decoupling capacitors next to connectors and vias that may require them
- Check TVS diodes or other ESD mitigation component pads are in series with the track to the component (i.e., any ESD event must pass through the component pad before getting to a sensitive device)
- Ensure there are no track stubs/net antennas going to test points or unused connector pins
- Ensure high-speed signals are routed as directly as possible—no scenic routes
- Check any track carrying over 100mA has its width calculated to ensure it is of sufficient size for the current
 - If the board will be fitted within an enclosure with little or no airflow, calculate or simulate the width for an internal layer instead of an external layer
- Check if an RF shield is required anywhere on the board and if it has a sufficient footprint
- If you are using a two-layer board:
 - Are there any ground loops?



Solutions for Every Role and Team

- Discover and evaluate design options earlier
- Develop in one shared workspace
- Scale with control and compliance
- Connect every stage of hardware development

[Learn More](#)

- Check panel fiducials are present
- Check for an origin identification mark
- Check impedance/layer/other test areas are present if required
- Check V Score/Milling/tab layers are present and aligned with the board if they are not in the board file
- Check the panel isn't too large for the board thickness and any milling (i.e., low flexibility/bounce)



Have something to add? Join the discussion.

Conclusion

If you are running through this PCB checklist or your own derivative created from it and using it in a PCB design review meeting, then it is best to consider each item as a discussion point rather than merely as a box to tick off. This can be really important to ensure the design meets the product's intent, rather than just ticking boxes. A review before production should be more than just ensuring that all the technical issues with the board you can consider having been addressed. It should also ensure the functionality meets the requirements. It's much more efficient to make any schematic changes and shift the electrical design to meet the design intent and requirements that may have been re-evaluated during the meeting than it is to just order the boards and then find out that you should have had that discussion before spending any money and needing to wait for the boards to arrive.



PCB Design Review Deep Dive
Altium Academy



Watch on YouTube

[Learn PCB Design Basics](#)

[Explore PCB Design Solutions](#)

[Getting Started Guide for PCB Design](#)

[PCB Design Documentation](#)



About Author

Mark Harris is an engineer's engineer, with over 16 years of diverse experience within the electronics industry, varying from aerospace and defense contracts to small product startups, hobbies and everything in between. Before moving to the United Kingdom, Mark was employed by one of the largest research organizations in Canada; every day brought a different project or challenge involving electronics, mechanics, and software. He also publishes the most extensive open source database library of components for Altium Designer called the Celestial Database Library. Mark has an affinity for open-source hardware and software and the innovative problem-solving required for the day-to-day challenges such projects offer. Electronics are passion; watching a product go from an idea to reality and start interacting with the world is a never-ending source of enjoyment.